INCH-POUND
MIL-PRF-20/27H
w/AMENDMENT 1
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SUPERSEDING
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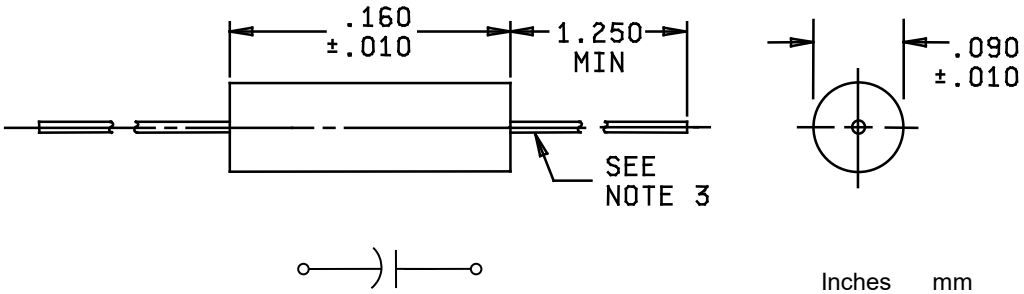
PERFORMANCE SPECIFICATION SHEET

CAPACITOR, FIXED, CERAMIC DIELECTRIC (TEMPERATURE COMPENSATING), ESTABLISHED AND NON-ESTABLISHED RELIABILITY, STYLES CCR75 AND CC75

CC75 is inactive for new design after 6 June 1984. Use CCR75.

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the capacitors described herein shall consist of this specification sheet and MIL-PRF-20.



CIRCUIT DIAGRAM

Inches	mm
.010	0.25
.090	2.29
.160	4.06
1.250	31.75

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Lead diameter shall be .016 inch (0.41 mm) +.005 inch, -.001 inch (+0.13, -0.03 mm).
4. Lead length may be a minimum of 1.00 inch (25.4 mm) long for use in tape and reel packaging, when specified in the ordering data.

FIGURE 1. Styles CCR75 and CC75 capacitors.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Lead type: Axial.

Case type: Tubular, molded.

DC rated voltage: See table I.

Operating temperature range: -55°C to +125°C.

Characteristic: CX, CK, CJ, CH, and CG in accordance with MIL-PRF-20.

Failure rate level (CCR75 only) (percent per 1,000 hours): M (1.0), P (0.1), R (.01) or S (.001).

Thermal shock and voltage conditioning (CCR75 only): In accordance with MIL-PRF-20.

Capacitance: Within tolerance specified (see table I).

Capacitance tolerance: See table I (B = ± 0.1 pF, C = ± 0.25 pF, D = ± 0.5 pF, F = ± 1 percent, G = ± 2 percent, or J = ± 5 percent).

TABLE I. Capacitor characteristics.

Part or Identifying Number (PIN) 1/ 2/ 3/	Rated voltage (volts, dc)	Nominal capacitance (pF)	Capacitance tolerance
CC - 75CXR10 - -	200	0.10	B
CC - 75CXR11 - -	200	0.11	B
CC - 75CXR12 - -	200	0.12	B
CC - 75CXR13 - -	200	0.13	B
CC - 75CXR15 - -	200	0.15	B
CC - 75CXR16 - -	200	0.16	B
CC - 75CXR18 - -	200	0.18	B
CC - 75CXR20 - -	200	0.20	B
CC - 75CXR22 - -	200	0.22	B
CC - 75CXR24 - -	200	0.24	B
CC - 75CXR27 - -	200	0.27	B
CC - 75CXR30 - -	200	0.30	B
CC - 75CXR33 - -	200	0.33	B
CC - 75CXR36 - -	200	0.36	B
CC - 75CXR39 - -	200	0.39	B
CC - 75CXR43 - -	200	0.43	B
CC - 75CXR47 - -	200	0.47	B
CC - 75CXR51 - -	200	0.51	B
CC - 75CXR56 - -	200	0.56	B
CC - 75CXR62 - -	200	0.62	B
CC - 75CXR68 - -	200	0.68	B
CC - 75CXR75 - -	200	0.75	B
CC - 75CXR82 - -	200	0.82	B
CC - 75CXR91 - -	200	0.91	B

See footnotes at end of table.

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TABLE I. Capacitor characteristics - Continued.

PIN 1/ 2/ 3/	Rated voltage (volts, dc)	Nominal capacitance (pF)	Capacitance tolerance
CC - 75CX1R0 - -	200	1.0	BC
CC - 75CX1R1 - -	200	1.1	BC
CC - 75CX1R2 - -	200	1.2	BC
CC - 75CX1R3 - -	200	1.3	BC
CC - 75CX1R5 - -	200	1.5	BC
CC - 75CX1R6 - -	200	1.6	BC
CC - 75CX1R8 - -	200	1.8	BC
CC - 75CX2R0 - -	200	2.0	BC
CC - 75CK2R2 - -	200	2.2	BC
CC - 75CK2R4 - -	200	2.4	BC
CC - 75CK2R7 - -	200	2.7	BCD
CC - 75CK3R0 - -	200	3.0	BCD
CC - 75CK3R3 - -	200	3.3	BCD
CC - 75CK3R6 - -	200	3.6	BCD
CC - 75CK3R9 - -	200	3.9	BCD
CC - 75CJ4R3 - -	200	4.3	BCD
CC - 75CJ4R7 - -	200	4.7	BCD
CC - 75CJ5R1 - -	200	5.1	BCD
CC - 75CJ5R6 - -	200	5.6	BCD
CC - 75CJ6R2 - -	200	6.2	BCD
CC - 75CJ6R8 - -	200	6.8	BCD
CC - 75CJ7R5 - -	200	7.5	BCD
CC - 75CH8R2 - -	200	8.2	BCD
CC - 75CH9R1 - -	200	9.1	BCD
CC - 75CH100 - -	200	10	FGJ
CC - 75CH110 - -	200	11	FGJ
CC - 75CH120 - -	200	12	FGJ
CC - 75CH130 - -	200	13	FGJ
CC - 75CH150 - -	200	15	FGJ
CC - 75CH160 - -	200	16	FGJ
CC - 75CH180 - -	200	18	FGJ
CC - 75CG200 - -	200	20	FGJ
CC - 75CG220 - -	200	22	FGJ
CC - 75CG240 - -	200	24	FGJ
CC - 75CG270 - -	200	27	FGJ
CC - 75CG300 - -	200	30	FGJ
CC - 75CG330 - -	200	33	FGJ
CC - 75CG360 - -	200	36	FGJ
CC - 75CG390 - -	200	39	FGJ
CC - 75CG430 - -	200	43	FGJ
CC - 75CG470 - -	200	47	FGJ
CC - 75CG510 - -	200	51	FGJ
CC - 75CG560 - -	200	56	FGJ
CC - 75CG620 - -	200	62	FGJ
CC - 75CG680 - -	200	68	FGJ
CC - 75CG750 - -	200	75	FGJ
CC - 75CG820 - -	100	82	FGJ
CC - 75CG910 - -	100	91	FGJ
CC - 75CG101 - -	100	100	FGJ
CC - 75CG111 - -	100	110	FGJ

See footnotes at end of table.

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TABLE I. Capacitor characteristics - Continued.

PIN 1/ 2/ 3/	Rated voltage (volts, dc)	Nominal capacitance (pF)	Capacitance tolerance
CC - 75CG121 - -	100	120	FGJ
CC - 75CG131 - -	100	130	FGJ
CC - 75CG151 - -	100	150	FGJ
CC - 75CG161 - -	100	160	FGJ
CC - 75CG181 - -	100	180	FGJ
CC - 75CG201 - -	100	200	FGJ
CC - 75CG221 - -	100	220	FGJ
CC - 75CG241 - -	100	240	FGJ
CC - 75CG271 - -	50	270	FGJ
CC - 75CG301 - -	50	300	FGJ
CC - 75CG331 - -	50	330	FGJ
CC - 75CG361 - -	50	360	FGJ
CC - 75CG391 - -	50	390	FGJ
CC - 75CG431 - -	50	430	FGJ
CC - 75CG471 - -	50	470	FGJ
CC - 75CG511 - -	50	510	FGJ
CC - 75CG561 - -	50	560	FGJ
CC - 75CG621 - -	50	620	FGJ
CC - 75CG681 - -	50	680	FGJ

1/ Complete PIN will include the following:
1st dash - Symbol “R” (for style CCR75) or dash will be deleted (for style CC75).
2nd dash - Applicable capacitance tolerance symbol.
3rd dash - Applicable failure rate level symbol (CCR75 only) or dash will be deleted (for style CC75).

2/ With respect to characteristics in previous revisions of this specification sheet: For capacitance values of 2 pF or less, characteristic CX is interchangeable with characteristics CG and CH; for capacitance values of 2.2 pF through 3.9 pF, characteristic CK is interchangeable with characteristic CH; for capacitance values of 4.3 pF through 7.5 pF, characteristic CJ is interchangeable with characteristic CH; and for capacitance values of 10 pF through 18 pF, characteristic CH is interchangeable with characteristic CG. These parts shall be stocked under the same applicable NSN’s.

3/ Temperature coefficient is not practically measurable for characteristic CX capacitors.

Temperature coefficient and capacitance drift: In accordance with MIL-PRF-20. For capacitors with values of 18 pF or less, the temperature coefficient tolerance shall be determined in accordance with table II.

TABLE II. Temperature coefficient tolerances.

Permissible capacitance change from capacitance at +25°C in ppm/°C					
Temperature	Characteristic				
	CX	CK	CJ	CH	CG
+125°C	1/	±250 ppm/°C	±120 ppm/°C	±60 ppm/°C	±30 ppm/°C
-55°C 2/	1/	+246.25 -326.25	+116.25 -166.25	+55.00 -91.25	+27.50 -53.75

1/ Not practically measurable.
2/ The ppm/°C values for -55°C were calculated by dividing ppm by negative 80°C.

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Marking: Method I of MIL-STD-1285, as shown in the following example:

JCK - "J" brand (ER only) (J); and characteristic (CK).

2R2 - Capacitance value.

CM1 - Capacitance tolerance (C); FR level, CCR75 only (M); and year code (1 for 2011).

ATM - Lot code (A); and trademark (TM).

Referenced documents. In addition to MIL-PRF-20, this specification sheet references the following document:

MIL-STD-1285

Amendment notations. The margins of this specification sheet are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army - CR
Navy - EC
Air Force – 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5910-2020-014)

Review activities:
Navy - AS, MC, OS, SH

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